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TOPICAL REVIEW

Bridging Languages in Healthcare: A Comprehensive Review of Multilingual and Code-Switched Chatbot Interactions

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ABSTRACT The Growth of societies with multilingual populations and the expanded accessibility to the healthcare service digitally has led to an increasing demand. For intelligent conversational agents that can operate and manage interactions with complex languages. Especially in countries like India, there exists a common process of code-switching between English and the regional or local languages, where the monolingual system falls conventionally. The paper explores the healthcare chatbots in the current state of research and the developments in the response and detection of queries, which involve multilingual and code-switching. This review aims to analyze the detection and response in the existing approaches to multilingual and healthcare queries. Also, the objective is to identify strengths, limitations, and research gaps in the current methodology. We highlight the challenges produced by the variation in the linguistic, languages with low resources, and the ambiguity of the context. Our analysis highlights that the performance of transformer-based multilingual models remains limited in low-resource and highly code-mixed settings, whereas the robustness of the models is improved. The paper analyses the models that exist for the identification of languages, detection of code-switching at the token-level, multilingual embedding, and systems based on transformers. We further discuss evaluation benchmarks, moral considerations, with ethical factors to direct future works in the equity of healthcare enhancement through AI with inclusive language.

INDEX TERMS Code switching, healthcare chatbot, natural language processing (NLP), multilingual NLP, text identification.

I. INTRODUCTION

Multilingualism is a feature that defines about numerous societies, with humans regularly switch the languages in communication. Healthcare chatbots are progressively used for primary or initial diagnosis, mental health support, and management of chronic diseases. Although the greater part of the existing bots is developed with a focus towards monolingual, there exists limitations to the patient usability employing blend with other languages.

These systems not only elucidate inputs with mixed languages but in addition can respond with suitable linguistic and medical accuracy. The increasing role of digital health, surrounded by global challenges like healthcare disparities among multilingual individuals, is broadening, and that lets conversational agents access the essential components of healthcare. Nonetheless, the medical chatbots have been deployed predominantly for the English-speaking users, thereby excluding many multilingual populations. The exclusion creates an exceptional challenge to the intelligent systems for conversations. For example, a patient may ask,"Mujhe aaj morning se headache hai, kya solution hai?" (Hindi-English) meaning "I have headache since morning, so what would be the solution?" and wait for a response that is relevant and coherent. Also, isolating the wide-ranging multilingual communities, there exists a mismatch in the language, leading to communication breakdown and adverse outcomes. This review explores and examines the way of multilingual medical query chatbots that can break down the language barriers. It enables an

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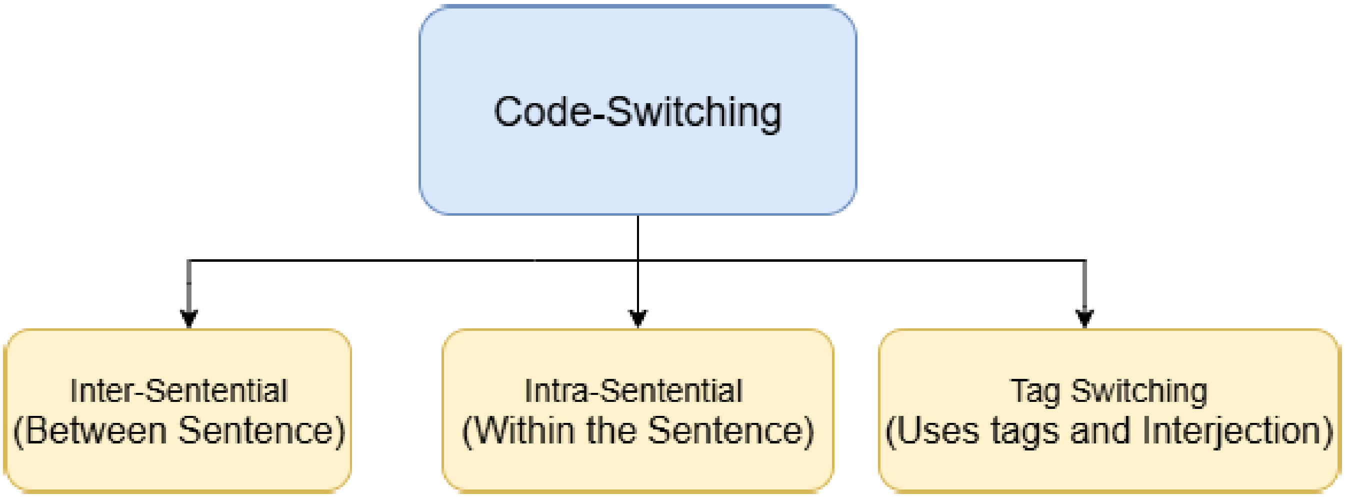

FIGURE 1. Classification of code-switching.

TABLE 1. Healthcare chatbot with systematic review methodology for Core NLP components.

NLP Component	Search Terms	Time Window	Inclusion Criteria	Reproducability Measures
Recognition of Intents	IEEE Xplore, ACM, Scopus, PubMed	2015–2024	Detection of Intent or intent tracking models in healthcare chatbots	Search strings, screening, and database steps documented
Named Entity Recognition (NER)	medical NER, clinical entity extraction	2015–2024	Extraction of medical entities (disease, symptoms, and drugs)	Criteria for explicit inclusion and a defined selection process
Dialogue Management	management of medical dialogue	2015–2024	Frameworks of multi-turn conversations for clinical interaction	Replicable selection protocol and specified scope
Response Generation	medical response generation, conversational AI	2015–2024	Sytems with Safety-Constrained and Context-aware medical response	Specified Sources and filtering methodology
Multilingual / Code-Switching	multilingual NLP, code-switching healthcare	2015–2024	Handling healthcare dialogues with Cross-lingual or mixed-language	Clearly stated Keywords, scope, and review boundaries
Overall Review Scope	Citation chaining and combined keyword search	2015–2024	Research based on Peer-reviewed NLP	Independent replication supported by methodology

clear, inclusive communication in the healthcare that reaches the people across various language backgrounds more effectively [1]. Although, developing multilingual healthcare chatbots effectively offers unique challenges which includes language uncertainty, cultural variations, coordination of clinical terminologies and the necessity for precise medical answers [2]. The dimensional analytics of the review over the Multilingual Healthcare chatbots has the following analysis:

- Healthcare, with the inclusion of Natural Language Processing (NLP), has significantly transformed the patient's approach to access, retrieve medical information, and promote health communication.
- To overcome the language barriers in global healthcare system, multilingual chatbots in medical queries have become a valuable tool, most precisely, also offering more effective communication and access to the multicultural patient communities [3].
- The systems supporting multiple languages permit patients to communicate in their native languages, reduce miscommunication, and promote literacy and health. The motivation derived from the reviews under-pinning the following contributions:
- This review explores and examines the recent advancements in managing queries with multilingual and code-switched with the healthcare chatbots.
- It structures and categorizes the different methodologies utilized in these area.
- The review highlights the challenges and current gaps in existing approaches.
- It recommends potential improvements aimed at enhancement of effectiveness in the real world and performance.

A systematic search was conducted using IEEE Xplore, SpringerLink, ACL Anthology, and Google Scholar. Studies published from 2006 to 2025 that identified multilingual and code-switching chatbots were considered. Out of approximately 410 screened articles, 70 were retained for comprehensive analysis that are depicted in Table 1. Existing surveys have explored healthcare chatbots and conversational agents, focusing on architecture, clinical applications, readiness, and deployment considerations [4], [5]. Separate reviews investigated multilingual NLP techniques and dialogue management, while the studies in recent times have analyzed code-mixed language processing in non-healthcare domains [6]. In contrast, this review provides a comprehensive analysis of multilingual and code-switched patient queries in healthcare chatbots by integrating NLP techniques with specific dialogue requirements, grounded in recent studies and real-world deployments.

II. METHODOLOGY OVERVIEW

A. UNDERSTANDING CODE-SWITCHING IN HEALTHCARE

Multilingual medical query chatbots have undergone an advanced and significant evolution of the methodologies, also progressed from simplistic rule-based techniques to sophisticated transformer-based frameworks. Initial models primarily depended on multilingual frameworks of rule-based, where the pattern-matching rules of handcrafted and the dictionary translation of keywords were utilized to analyze queries of patients in various languages [7], [8]. However, these systems lacked in the scalability and adaptability over the medical language which are complex and ambiguous and on the other side these systems were effective in terms of prediction and interactions based on the symptoms. The subsequent beneficial evolution occurred with the initiation of Statistical Machine Translation (SMT) techniques, notably phrase-based SMT frameworks like Moses, which involved in the translation of queries from patients into a pivot language (commonly English) in advance of processing them to a pipeline with monolingual medical NLP [9]. Although SMT improved and refined the chatbot's ability to handle various languages as cross-lingual coverage, still recurrent mistakes in the translation frequently diminished the diagnostic reliability and the accuracy of its response. The performance of SMT is highly sensitive to the structure of code-switching, despite extending the cross-lingual accessibility. Finally, here is a description of different domains used in the various areas and the unexplored areas as a comparison represented in Table 2. Inter-sentential code-switching performs an effective translation due to sentence-level independence, whereas intra-sentential code-switching involves multiple languages within the same sentence, posing significant challenges for the SMT models. The classification of Code Switching into Inter-Sentential, Intra-Sentential, and Tag Switching is represented in Figure 1 which also individually describes on its types.

B. RELEVANCE IN HEALTHCARE

In the context of medicine, the symptoms are described by the patients in their own native languages, including the medical terms in English. To describe this, namely a patient says "Kal se mujhe chest pain ho raha hai (meaning "Since yesterday I have chest pain") and it's becoming more severe". To interpret such sentences, NLP systems are required, and that also overreaches the basic translation and requires cultural, linguistic, and medical understanding. We performed a comprehensive systematic literature review emphasizing the multilingual and medical query chatbot systems. The accelerated advancement of NMT (Neural Machine Translation), enabled by architectures of sequence-to-sequence incorporating attention mechanisms, modified the design of multilingual chatbot supporting translation with more context-awareness and with fluency [15]. In the context of Healthcare, the translation of patient queries into a language-oriented representation for medical intent classification is employed by NMT systems; subsequently, responses are re-translated into the user's desired language [16]. In recent times, multilingual language models based on the transformers, namely multilingual Bidirectional Encoder Representation from Transformers (mBERT) [17], Cross-lingual Language Model - Roberta (XLM-R) [18], Multilingual Text to Text Transfer Transformer (mT5) [19], [20], and domain-oriented variants such as BioBERT-multi manifested as the foundation of medical chatbot systems in multilingual models. These models exploit vocabularies with subword shared and multilingual vector representations, permitting a Monolithic model to facilitate various languages, frequently achieving transfers with zero-shot or few-shot to languages with low resources [21].

Concurrently, the pretraining model for domain-specific multilingual had a surge, in which the biomedical corpora in numerous languages were utilized to optimize the base models, substantially strengthening their ability to interpret the medical terminology with complexity. The Vital research focus of countries as India and the Philippines is handling code-switching, which is a general phenomenon in conversations between patients. To the course of action for the queries with mixed languages like "fever aur headache jab se kal" [22], (which means "I've headache and a fever since yesterday"), the modern models utilize the language identification at the token-level fused with sensitive context mappings. Although the detection of token-level code-switching improves fine-grained understanding, it increases the computational costs overhead stemming from the identification of per-token language, longer subword sequences, and an elevation in the complexity of self-attention. This overhead can elevate inference latency, limiting the performance of the chatbot in real-time. To mitigate this, hybrid strategies are relied upon, which integrate sentence-level detection with token-level refinement. As well as additionally, the incorporation of speech-to-text systems such as multilingual Automatic Speech Recognition(ASR) and Whisper models facilitates interactions based on voice, which are crucial in populations with levels of low literacy [23]. The Implementation of integrating multilingual and code-switched ASR in voice-based healthcare chatbots can be done using shared acoustic models combined with language-aware decoding mechanisms. In which the concatenated tokenizers allow token-level identification of language while reusing vocabularies with monolingual subword units [24]. The robustness

TABLE 2. Overview of domain, focus and gaps in code-switching.

Domain	Areas Covered	Areas Unexplored
Computational NLP [10]	CS Synthesis, Simulation, Identification, and also Enhancement	Sociocultural linguistic traits, limited resource multilingual data
Cognitive Psychology [11]	Cognitive regulation, task-switching costs, comprehension and verbalization	Neuroimaging in natural contexts, long term cognitive impacts
Child Language Development [10]	Usage-based foundations of Code-Switching in early speech	Prolonged longitudinal studies on childhood development
Sociolinguistics [12]	Purpose of Code Switching in media and education	Age-related discourse variation, identity/emotion-driven bilingual switching
Formal Linguistics [13]	Theoretical grammar restrictions and models	Global vs context-specific validity, emerging language pairs
Education [14]	Effectiveness of CS in education	Wide-ranging, comparative analysis across demographics and age cohorts

in low-resource settings is improved when the data sparsity is mitigated by synthetic code-switched speech generation and data augmentation. Typical ASR error patterns comprise rare and technical terms misrecognition, dominant-language bias, and language-switch boundary errors are observed frequently [25]. These can be mitigated by using strategies such as medical lexicons, constrained decoding strategies, post-ASR correction, human escalation mechanisms, and confidence-based clarification strategies. These systems transform the patient queries in spoken form into text, process them through the pipelines of NLP, and return back to the responses spoken in the language targeted. The current systems with more advancements make use of the architectures framed as hybrid that consolidate NMT, Multilingual embedding and domain-focused graphs with medical knowledge to improve the relevant context and accuracy [26]. The Components involved in Multilingual environment based on the Healthcare Chatbots are described in Figure 2. For instance, a query of a patient about the shortness of breath and chest pain in Swahili can be translated, XLM-R utilized by embedding with semantics and incorporated to corresponding clinical protocols stored in a graph with multilingual knowledge prior to producing an medical response which is appropriate [27]. In collaboration, these methodologies exemplify the progression technologically rule-based, rigid systems to adaptable, insightful and context-aware multilingual NLP solutions, signifying a major milestone in the accessibility of global healthcare.

C. CORE NLP COMPONENTS

The Pipelines of a typical chatbot includes input text, detection of language, tokenization, intent classification, recognition of entity, generation of response and rendering the output. Identification of languages and mapping the cross-lingual are additionally required by the multilingual systems [28]. The Components of Core NLP collectively facilitate the process inputs from patients across various languages with high accuracy in the multilingual medical query chatbots. The initial step involves detection of language and identification, which establishes the language and detects the code-switching at the token or the sentence-level. This is commonly achieved in implementation using light-weight language identification models (for instance, transformer-based token level predictors or fast-text based classifiers) which detect code-switching patterns before routing the query to the correct processing pipeline. In healthcare settings, these remain critical since incorrect language detection may generate errors further down the line.

On subsequent of it, is the preprocessing of text and normalization, in which the input is undergone with the process of tokenization, lemmatization, and cleaning. Cleaning process involves, elimination of noise like errors in the spelling, slang, or abbreviations which are non-standard. And at the same time, converting of colloquial expressions (e.g., "BP high", "sugar problem") into medical terms by frequently incorporating synonym dictionaries and rule-based medical terminologies is crucial. To further support effective handing of multilingual inputs and low-resource language processing, subword tokenization techniques (SentencePiece or WordPiece) are employed.

Subsequently, the normalized text is converted into dense vector representations by multilingual embedding generation. The multi-lingual embedding models, such as mBERT, XLM-R [29], or embeddings of specific areas to map terms that are semantically equivalent across various languages into a space with shared representations that are employed by advanced systems.

Once the text is depicted clearly and meaningfully, NER [30] recognizes and categorizes the vital medical entities such as diseases, symptoms, prescriptions, and dosages. NER is implemented typically using fine-tuned transformer-based sequence-labelling models often utilizing ontologies like Unified Medical Language System(UMLS) or Systematized Nomenclature of Medicine – Clinical Terms(SNOMED CT) [31]. Concurrently, detection of intents and models of classification determines the benefits of the query for instance,whether it's checking the symptoms, scheduling the appointments, clarification of prescription or intent of any other healthcare related queries. These models are typically implemented as supervised intent classifiers operating on multilingual embeddings thereby supporting effective intent recognition under zero-shot and few-shot language settings. The literal translations are avoided which could lead to errors clinically for understanding the cross-lingual queries and the medical context awareness of machine translation systems confirms that the meanings of domain specific areas are preserved.

The context of the conversations, flow of interaction and the interfaces related to the sources of medical knowledge are then maintained by the dialogue management module of chatbot. The system then connects medical knowledge bases, resources related to clinical protocols and decision-support to resolve relevant medical information. To ensure the accuracy, retrieval-based methods are integrated with generative language models as a hybrid approach for response generation. The hybrid approach provides fluent and an empathetic interaction. The foundation of all these is the integration of basic medical knowledge, connecting the structured medical databases with the chatbot, knowledge graphs, and clinical updates in real-time. The medical information which is ambiguous or harmful are filtered, and all responses sound medically, ethically responsible, and complaint with regulations that ensure the safety and compliance layer [32].

D. MODELS OF MULTILINGUAL NLP

Across 100+ languages which use vocabularies with shared subword and pretains cross-lingual be enable processing the transformer models - mBERT, XLM-RoBERTa, and NLLB. The chatbot in the healthcare are designed to process multiple language queries which involves linguistic gaps bridging and enables the medical assistance inclusively shape a backbone with the multilingual NLP models. To access the healthcare information accurately, these models are developed to manage both high-resource and low-resource languages. These languages assure a diverse linguistic background and patients approach the healthcare information. A shared semantic space is provided by advanced architectures like mT5, XLM-RoBERTa [33], multilingual BERT (mBERT) to align the similar expressions from different languages and allows understanding the cross-lingual without the requirement for the parallel training data in each case. In the implementation of healthcare, these models are frequently modified on the corpora of domain-specific obtained from the medical literature, clinical notes, and forums of patients, facilitating them to understand on a expert vocabulary and jargon. A critical enhancement is the consolidation of context-aware translation mechanisms that are responsible for medical nuances, decreasing the risk in scenarios that are sensitive with mistranslation and scenarios such as reporting the symptom or instructions of the dosages. Systems also employ transfer learning based on adapters and techniques. Knowledge distillation is involved in improving the low-resource languages and its performance and also providing that even rare languages or idioms receive necessary coverage. The incorporation of electronic health records and Biomedical Ontologies improves the sensitivity of the model(bias, robustness and fairness of NLP models), which also creates complexity to the medical contexts. Task-level accuracy is improved by fine-tuning, but it increases the computational complexity. Also, fine-tuning causes the risk of catastrophic interference when multiple languages are adopted by the model simultaneously. Adapter-based methods reduce training time and memory consumption. It enables efficient adaptation to low-resource languages but may underperform fine-tuning in terms of maximum accuracy. Large multilingual models are compressed into smaller ones using knowledge distillation. While knowledge distillation supports fast, low-cost deployment through model compression, but potentially limits the performance on complex medical queries. With these strategies in healthcare environments, the trade-off between operational feasibility and the accuracy of the model is illustrated.

For the applications in real-time, these models frequently influence the capabilities of learning the zero-shot and few-shot, empowering them to establish to new languages or terminologies in medical field with fewest labeled data. Further, multi modal integration with advancements allows models to proceed with not only inputs like textual even images, speech and prescriptions in the form of scanned are the crucial in cases when symptoms couldn't be typed by patients. Moreover to enhance the reliability, mechanisms handling code-switching are inscribed, facilitating uninterrupted interpretation of the inputs with mixed-languages, which are general in the multilingual societies [34]. In addition, using the patient data with decentralization and without privacy compromise, the models can be trained and revised which is ensured by the deployment of frameworks in federated learning. The Pipelines with evaluations of continuous model along with the review mechanisms of human-in-the-loop ensures the monitoring of multilingual performance, detection of biases, and the error pattern that concern the language-specific are rectified and improved over time [35]. Eventually, the NLP models which can support multilingual empowers the chatbots in healthcare to provide equitable, contextually precise and medical support with cultural sensitivity(linguistic and sociocultural appropriateness), align with the goals of global health accessibility. The MLM is the core objective used in NLP like BERT in which certain words in a sentence are masked and the predictions to be learned from the model. Here is Table 3 describing the comparative analysis of MLM over various languages and the typical applications with their key features. Also, gives information about the number of languages covered with each model in various applications.

E. ADAPTATIONS OF MEDICAL NLP

Models with specialization, such as BioBERT, Clinical-BERT, and Variants, recognize the terminology based on

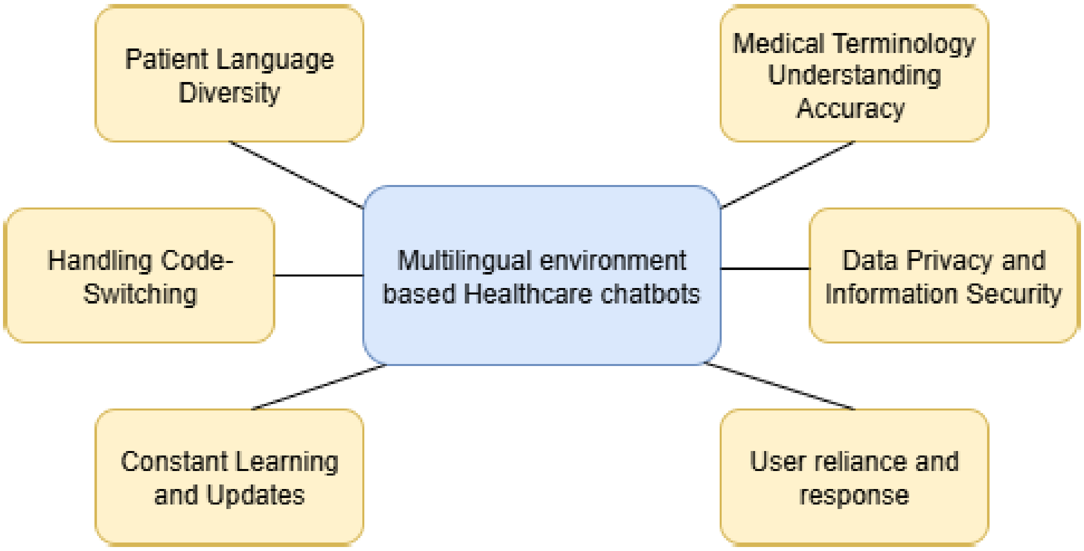

FIGURE 2. Component in multilingual based healthcare chatbots.

TABLE 3. Comparative overview of MLM (multilingual language models).

Model	Language Coverage	Key Features	Typical Applications
mBERT [17]	104 languages	Based on Transformers, Multilingual based Masked Language Modeling(MLM)	Text classification, Named Entity Recognition (NER), QA
XLM-R [36]	100 languages	RoBERTa pretrained with Robust Multilingual	Multilingual understanding, Tasks with text
mT5 [37]	101+ languages	Text-to-text sequential Transformer, Paraphrasing - Generative capabilities	Machine Translation, Context Summarization, Language Generation
Cross-Lingual Speech Recognition (XLSR) [38]	Many speech languages	Multilingual speech embeddings	Multilingual Speech Recognition(MSR)
Many to Many (M2M)-100 [39]	100 languages	Non-pivot based Multilingual Translation(Many-to - Many)	Direct multilingual machine translation

domain-specific by training on the biomedical corpora, frequently combining with SNOMED CT, UMLS, or ICD-10 for consistent understanding. The NLP systems with tuning of specialization are for understanding [40], interpreting, and responding accurately to queries related to healthcare, completely involved in the adaptations of Medical NLP. A Pre-training of domain-specific over biomedical datasets(in large-scale) like PunMed, MIMIC-III [41] and reports of clinical trials for capturing the terminologies with specialization, abbreviations and expressions related to diagnosis which are unlike the NLP models for general purpose. With the help of resources namely UMLS, ICD-10 and SNOMED CT, these models are frequently enhanced with the integration of ontology driven based knowledge to assure the constant interpretation of diseases and its symptoms and treatment variants in various languages. The Modules with disambiguation which involve context sensitivity(dialogue continuity, situational awareness) are in inclusion which can differentiate between the various meanings of common terms in the context of medicine is another vital critical adaptation, in real-time we may use the word "Cold" described as temperature state vs. disease in human.

Moreover, the environment supporting Multilinguistic, the NLP systems in medical domain assimilates the terminology in the medical field which are cross-lingual integration to assure the concepts of the healthcare with equivalence are guarded at the time of translation and controlling the linguistic drift to the minimum. NER models are refined in specific for the healthcare entities like the names of the drug, locations of the anatomy, result from the lab and the terms supporting the procedures, facilitating the information of the patients which are extracted precisely [42]. Several systems with advancements are embedded with the mechanism, supporting temporal reasoning to interpret the symptoms of timelines, schedules of the medications and the progression in the disease. These mechanisms are vital components to diagnose accurately and provide the recommendations. The Chatbot can highlight the attention of the patient cases that is needed with the intervention of human and medical rather than depending solely on the responses that are automated, which can be done by implementing the confidence scoring to ensure the estimation of uncertainty and safety. Together with, strategies with bias mitigation are achieved in the way of utilizing sampling with balanced datasets and training the model with the fairness-aware, despite are crucial for various demographic or linguistic groups to prohibit the medical advice with disparities. The real-time medical recommendations which are personalized, ensures the regulation compliance namely Health Insurance Portability and Accountability Act(HIPAA) and General Data Protection Regulation(GDPR) [43] are enabled through the APIs [44] that are secured in integration with EHRs (Electronic Health Records). Few systems also utilize the multi modal NLP in medical field to systematize and associate the symptoms in the form of textual descriptions with the images from radiology, results from lab and sensor data those are wearable, which creates a assessment of the patient to be holistic. Finally, the loops in continuation in the feedback with clinical people permits these models to emerge the medical knowledge together, assuring the recommendations of the chatbots stay stable based on the evidence and are aligned with the recent guidelines of the treatment.

III. BACKGROUND AND THEORETICAL FOUNDATIONS

A. DIALOGUE MANAGEMENT

Dialogue Management is fundamental to a healthcare chatbots system, as it directly impacts the safety of patients, diagnostic reliability, and compliance with regulatory standards. In contrast, to open-domain conversational agents, healthcare chatbots are supposed to handle safety-critical interactions, multi-turns in which inconsistencies can lead to consequences clinically. The role of dialogue management is to ensure safe, multilingual, and context-aware operations in chatbots. The generative strategies, retrieval-based, and rule based are adopted by the healthcare systems [45]. In combination of the retrieval with the generation, that is, for the accuracy with the fluency has been considered commonly increasing healthcare applications. For a healthcare chatbot, the management of the dialogue is considered to be the central control mechanism coordinating the conversation flow in between the system and the user and alongside ensuring the relevance in the contextual, safety and the responses that are coherent. This component should manage the continuity in cross-lingual, despite managing the track of context across the numerous turns which ensures the meaning and accuracy in medical terms are conserved when languages are switched in the middle of any conversation. The Dialogue Managers in the advanced systems often utilize the approaches that are hybrid which combines the frameworks of rule based models for the scenarios related to critical safety (for instance, red flag symptoms which refers to specific warning signs in a patient's condition) to be flexible with the machine learning models and handling the natural conversations [46]. The renewed representation of intents of the users, along with the symptoms mentioned and advice given in prior are maintained by the state tracking modules which is critical in the assessments of healthcare at multi-step. Specifically, for an urgent or serious threatening situation, the dialogue management has to determine when a human healthcare provider to be informed by an escalation of the query from the user. So, it incorporates the mechanism of prioritization of intents. Reinforcement Learning(RL) is utilized by the advanced systems to upgrade the strategies of the dialogues over time, by learning from the outcomes of the interactions for the betterment of the satisfaction, accuracy of the diagnosis and the efficiency of the patients [47]. In the context of the healthcare, the mechanism involving the recovery of error are importantly specific, the chatbot identifies the misunderstandings, clarifications of requests and revise its course without interrupting the flow of the conversations. The protocols related to safety can also incorporate the gating in the conversation, in which the delicate topics like the idealization of suicides, complications in pregnancy or any reactions related to allergy with severity can trigger the predetermined juncture of the workflows.

There are many challenges faced by the Multilingual NLP which is explored in Table 4 with their detection and response challenges. The dialogue managers can incorporate with the profiles of the users and EHRs(Electronic Health Records) [62], facilitating the context-aware and specified for the patient recommendations. At the same time respecting the regulations of the privacy such as HIPAA and GDPR [63], to improve the personalization. Analysis of Sentiment and the emotions are progressively incorporated into the management of dialogues to identify the anxiety of the patient, distress or frustrations, permitting the system to react empathetically and customize the tones respectively. In the multilingual environment, the strategies with cultural adaptation can be integrated with dialogue managers whereas the phrase, metaphors and the norms of conversations are customized to the cultural background of the patient to improve the engagement and the trust in the multilingual environments. Likewise, few dialogue systems in the healthcare with the advancements support interactions of the multi-parties, facilitates the doctor, patient and the chatbot to concertedly work together in the consistent thread of the conversations [64]. As a conclusion, uninterrupted steady monitoring, auditing with quality assurance and the human in the loop of clinical validation assures that the management of the dialogues abide accurate, safe and medically aligned along with the standards of the healthcare [65].

IV. CHALLENGES IN DETECTION AND RESPONDING TO MULTILINGUAL QUERIES

The Various challenges in the detection and response to the query generated in multilingual model are demonstrated in Figure 3.

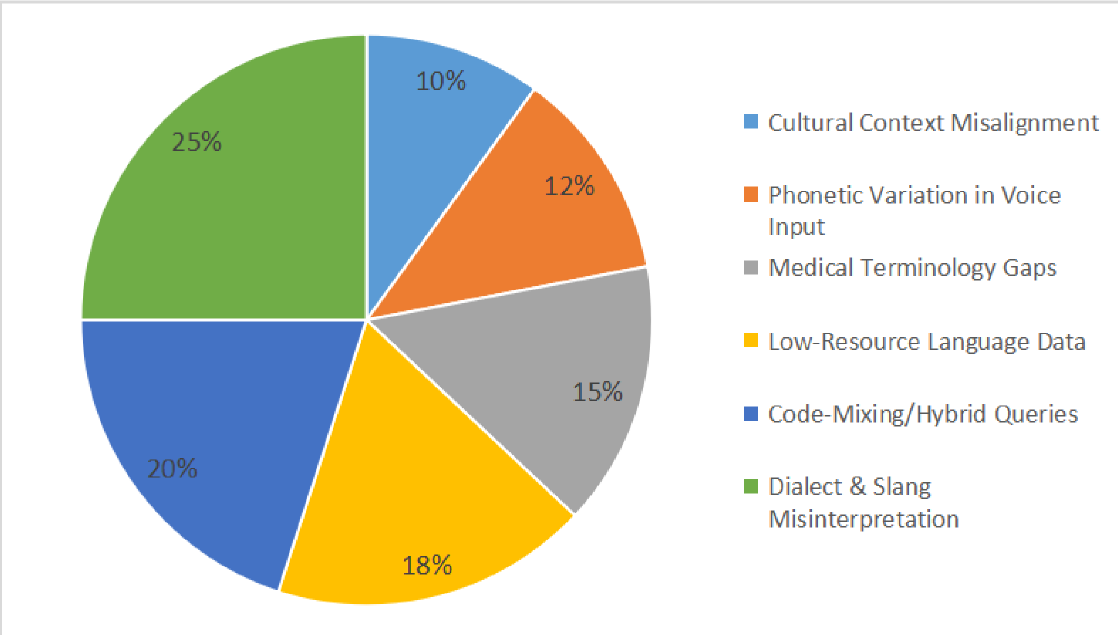

FIGURE 3. Challenges in detection and responding to multilingual queries.

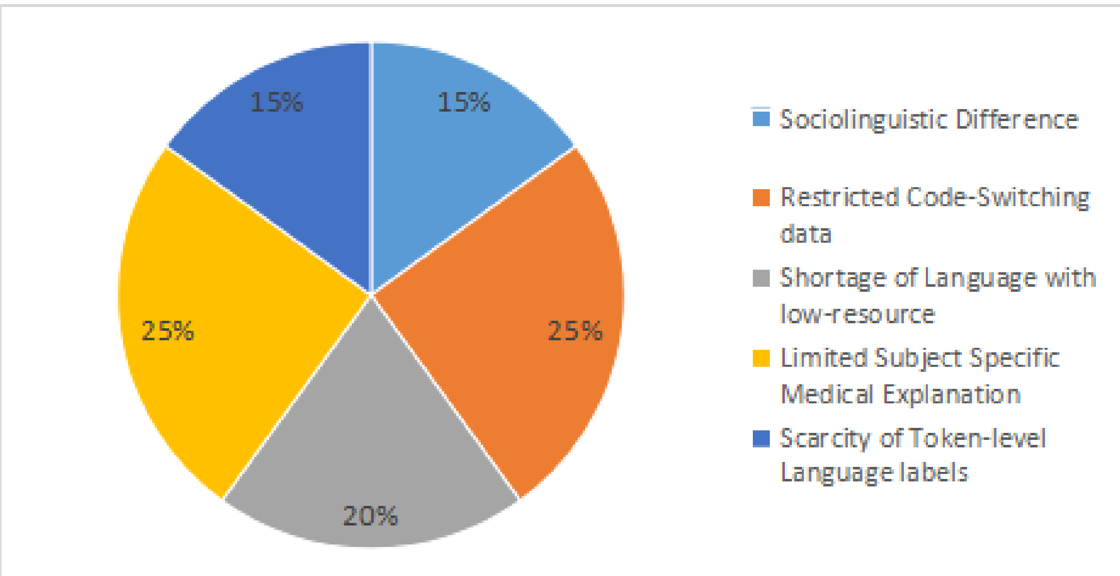

FIGURE 4. Challenges in lack of illustrated dataset.

A. AMBIGUITY AND CONTEXT LACKING

The queries which are code-switched frequently lack in the structure of the grammar and heavily rely on the context [66]. Words from various languages might have the meanings and spelling overlaps, which could lead to ambiguous interpretations.

A Healthcare Chatbot with the multilingual model are launched in a clinic which serves public and aims to provide assistance to the patients in the context of responding to the queries related to the healthcare in English, Arabic and Spanish. Although a crucial incident happened when a user who spoke spanish provided a input, ‘‘Yoh tengho dolar en el pecho y me seintuh un pechoo frio’’ which meas I have chest pain and I feel a bit cold, was misunderstood by the chatbot. The chatbot considered and remedied the ‘‘frio’’ as a symptom of a usual cold instead of a sensational symptom, failing to notice or even ignoring the serious concurrent symptom as chest pain. This kind of lack in the

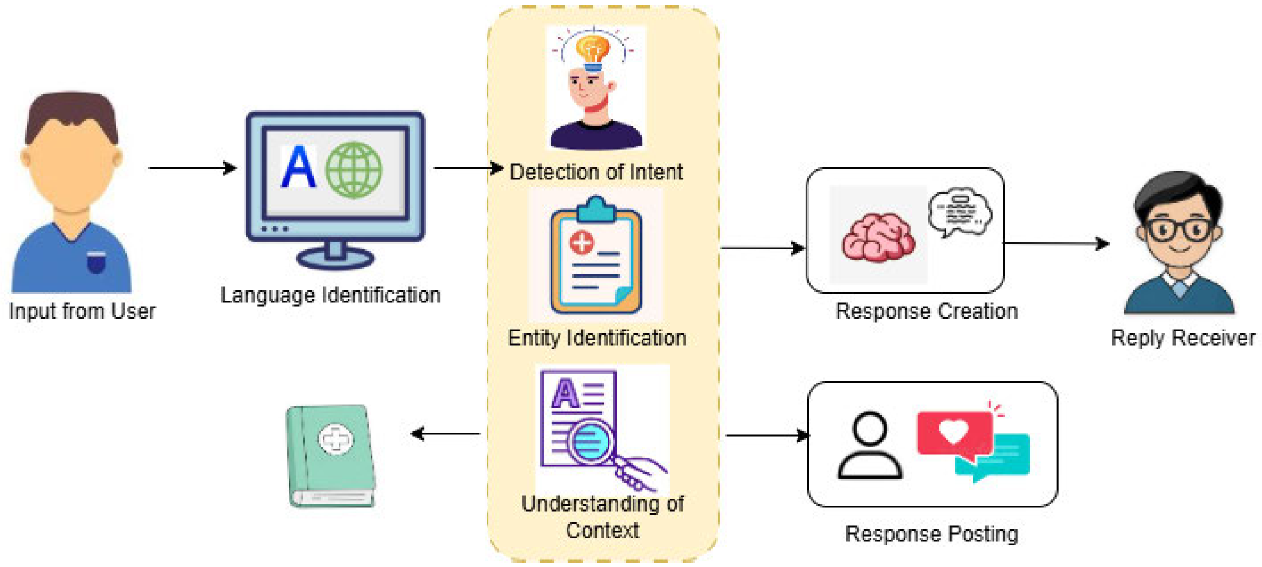

FIGURE 5. **Workflow of response in multilingual chatbot.**

understandings of the context and the disambiguate of the failure for the term led the chatbot to prescribe or advice to rest and intake fluids rather than for an emergency of immediate attention needed medically. The design of the chatbot was highlighted with the crucial shortcomings or flaws, specifically to handle the ambiguity in linguistic [67] and analyze the severity of the symptom in various languages. The underlying causes includes over dependency of the translation, unavailability of contextual reasoning, and inadequate assessing of urgency in the protocols of healthcare. Finally, the case study highlights the benefit for the medical NLP systems that support multilingual which can integrate the understanding of the native-language, context awareness in terms of medicines and precautions which are built-in; such as clarification of the prompts and escalation of human for preventing prospective outcomes which are also harmful.

B. LACK OF ILLUSTRATED DATASETS

In the medical domain, especially languages with low resources suffer an insufficiency of labeled datasets, whereas the current corpora are inadequate for preparing models with robustness. To develop a healthcare chatbot supporting Multilingual content is a major challenge, as the annotated datasets are considered to be lacking in their sufficiency [68], specifically for languages that are low-resource and code-switched. For example, imagine a chatbot related to healthcare is designed to assist patients belonging to India who often or subsequently use English with their local or regional languages, such as Tamil, Telugu, or Hindi, to describe their symptoms. Whereas the medical corpora of English are abundant, parallel or datasets annotated are used to capture the queries with nuances in the multilingual patients. For instance, ''Doctor, kal raat se mujhe chest mein pain ho raha hai'' is with scarcity (meaning ''Doctor, since yesterday night I have been experiencing chest pain'').

The Chatbot struggles to detect the intent accurately, symptoms identification or next steps recommendation without the labeled data [69] and these could compromise the safety of the patient. Developers frequently enforced to depend on transfer learning from the languages with high resources or from the synthetic data utilizing the machine translation. These approaches face failure to capture the context of interactions with true conversations and cultural in multilingual healthcare. This kind of scarcity in the data slows down the training of the model however it also provides limitations to the chatbot in the performance related to the real-time scenarios, by spotlighting the crucial requirement for the initiatives in the creation of collaborative dataset that are focused on healthcare related multilingual NLP. The Figure 4 illustrates the challenges faced if there exist a lack of illustrated dataset.

C. IDENTIFICATION OF TOKEN-LEVEL LANGUAGE

The language detection of every word in a sentence are on demand with the analysis based on fine-grain, specifically when the words are transliterated, borrowed, or typed with phonetics facilitating various scripts (for instance, Hindi written in English Script). There exists a crucial challenge in the healthcare chatbot supporting multilingual which is the identification of languages at the token-level [70], which becomes a requirement when the various languages are mixed by the patients within one single sentence or query. For instance, a patient from India may type, ''Doctor, mujhe vomiting ho raha hai, kaal raat se'' which is a combination of Hindi and English(meaning ''Doctor I have vomiting since yesterday night''). In the Sentence level, the chatbot may incorrectly do classification of the entire input as Hindi or

TABLE 4. Challenges in multilingual NLP.

Category	Detection Challenges	Response Challenges
Language Identification [48]	Concise queries or code-switching induced language ID errors	Choice of relevant language or model for response
Script and Encoding [49]	Hybrid scripts, non-Latin characters parsing errors	Visualizing and creating output in relevant script/alphabet
Linguistic Diversity [50]	Fluctuating grammar, tokenization, segmentation (e.g., Chinese/Japanese)	Developing grammatically appropriate/idiomatic responses for every language
Low-resource Languages [51]	Confined training data for exceptional languages minimizes detection accuracy	Low-quality output, Elevated bias for Marginalized languages
Ambiguity and Polysemy [52]	Query denotation may rely on language/culture	Catering appropriate meaning-specific answers, especially for idioms
Code-switching and Mixing [53]	Detecting multiple languages using individual query	Creating code-switched or multi-language result logically
Bias and Imbalance [54]	Supremacy of languages with high-resource in models	Result may be Correct/risk distorting cultures
Technical Limitations [55]	Scalability: numerous models enhances infrastructure	Upholding system speed and accuracy across various languages
Contextual Understanding [56]	Identification of user intent with cultural/semantic nuances	Offering contextually relevant responses, not just Verbatim translations
Query Length and Ambiguity [49]	Very concise queries are rigorous to allocate a language, specifically for named entities or terms technically	Responses may need additional clarifying probes or fallback mechanisms to prevent misunderstanding
Tokenization and Morphology [57]	Languages without continuous (Chinese, Japanese, Thai) and complicated morphology (Arabic, German) complex segmentation and normalization	Responding needs appropriate inflections and idiomatic structures, which diverge substantially by language
Script and Encoding Complexities [58]	Hybrid scripts or encoding without standards can minimize Identification/classification accuracy, specifically in legacy data	Result generation needs robust context support and normalization, specifically in transliterated or hybrid-script cases
Data Scarcity and Imbalance [59]	Languages with low-resources have datasets restricted, provoking weaker identification and amplified errors	Output may enshrine bias or deficit coverage for marginalized languages, reducing inclusiveness
Named Entities and Localization [54]	Multilingual queries might utilize various forms of the similar named entity or place (e.g., "Mumbai" vs. "Bombay")	Producing context-appropriate, localized output is rigorous across identities and regions
Cultural and Semantic Nuance [60]	Misunderstanding queries with powerful cultural or contextual synonym (idioms, references) is regular	Producing culturally applicable and contextually reliable responses without initiating bias or errors
Evaluation Benchmark Gaps [61]	Not many multilingual, code-switched, or context-rich criterion result in blind spots in system validation	Deficiency of extensive test cases determinants inconsistencies and undetected failures in real-time use
Ranking and Relevance [55]	Integrating language-specific ranking elements (e.g., term frequency, phrase proximity) hinders retrieval	Holding suitability of high across languages and scripts, specifically for Hybrid language knowledge reclamation
Safety, Fairness and Risk [54]	Misclassification might cause to visibility to unsafe or irrelevant content in the incorrect language	Output may show or magnify language/cultural tendency and safety threat, particularly for marginalized groups

English, which results in misinterpretation. The system has to identify that "Doctor" and "Vomiting" are the tokens English language and other words are belonging to Hindi language, however at the token-level identification. This type of classification which are fine-grained is specifically a major part in the context of healthcare where the meaning precise in the terms of medical field namely "diabetes" or "fever", must be protected and maintained in English, however they appear with other languages surrounded by the words related to the contexts. The process has errors and which can lead to detection of symptoms that are missed or the inappropriate classification of intent, indeed influences and create a impact in the quality of care that a chatbot can deliver. The annotated dataset which are code-switched are required by the models that support the token-level identification and are built accurately, the datasets are also with language tags for every individual token, but those such resources face scarcity. Accordingly, the sub-word embedding and the heuristic rules are retreated by the developers, in which the sub-word embedding are not regularly robust in tackling the unpredictability of the real-time conversations of the multilingual patients. The below Figure 6 represents the percentile of the various aspects that cause the identification of token-level language as a challenge.

D. SUBJECT-SPECIFIC VOCABULARY

The terminology in the healthcare includes one another complex layer. For many specific conditions over medically, several languages lack the translation that can be taken up directly or even utilize the terms in the English language which are transliterated. There exist a various challenges that are related to nuances presented by the queries which are code-switched and process multilingual [71] however for building the healthcare chatbots with reliability. These hurdles are solely linguistic but connects with technical, ethical and social demographic domains perplexing with the development of the chatbot, specifically in the settings of healthcare in which the empathy and exact accuracy are dominant and major. The healthcare chatbot with the multilinguistic model also has one another critical challenge in handling the vocabulary based on subject-specific, in particular to the terminologies of the healthcare coincides with the language used everyday or difference across the various languages. For example, a patient may tell like, ''Mujhe sugar hai''(meaning ''I have sugar''), in hindi which conversationally indicate the diabetes instead of the meaning literally for ''sugar''. Equivalently, the local terms such as the Tamil word like ''Vayu'' can be utilized to explain the discomfort in the gastrointestinal, that has no explicit equivalence to English. The colloquial expressions and the terminology related to the medical field or healthcare are to be capable of distinguished by the chatbot for distributing the responses accurately related to the healthcare. Moreover, the corpora utilized for general purpose is been trained for the most language models with multilingual and doesn't achieve to snatch the usages based on the specialized healthcare causing misinterpretations. Figure 5 represents the workflow of response in the Multilingual Healthcare Chatbot. Then the code-switched queries with English medical terms namely ''BP'' or ''Cholesterol'' are enclosed within the local regional languages and this become even more complex. The Input from the patient under the critical stage faces the misunderstandings by the chatbots and fall in the risk without the right to approach the annotated datasets which can combine the vocabulary based on colloquial, regional and healthcare based. Narrowing down the mentioned gap requires combining the lexicons that are subject-specific, structuring the knowledge gaps based on the healthcare, and refined models oriented over the healthcare multilingual corpora to assure the accuracy and the interactions relevant to the culture [72]. Now the Figure 7 explains the various aspects that cause the challenge of subject-specific vocabulary as in represented with the percentile data.

E. VARIABILITY IN SOCIAL LINGUISTIC

In the multilingual NLP, one of the most unnoticed overlook is the patterns that are code-switched with the variation over the demo graphical regions. The purpose of code-switching can vary with their benefits among the rural and urban people [73] and also in between various groups with difference in age whereas we can also expect the variation in the structure, frequency even enclosed in the similar kind of pairs of languages like the Hindi-English. For instance, young generation group may do the code-switching for the reason of styling, whereas the groups with elder or older people can do switching between languages on account of level of comfort or the vocabulary gaps. The approaches for modeling and the data training models are integrated into the social linguistic diversity which persist as a significant constraint. There exists a considerable challenge presented by the sociolinguistic variation in terms of the multilinguistic healthcare chatbots with effectiveness, since the patients frequently converse their medical concerns in different ways which are impacted by their cultural, regional and social backgrounds. For instance, a patient who is young and from a urban area expresses his medical concern on headache by saying in English ''I have a migraine from morning'' while an elderly speaker from a rural area may describe it in Hindi ''Sir, subah se sir dard lag raha hai'', which literally means ''Sir my head has been hurting since morning''. There remains a similar meaning in both the phrases which conveys about the severe headache, but the variations in the sociolinguistic style, inclusion of the difference in the phrasing, formality levels and the language used metaphonical create a challenge for the chatbot to understand and interpret them in the similar accuracy. There is another layer of complexity existing for the code-switching, since the alternation of languages occur among patients and tones enclosed in the same message to describe the critical situations or politeness or the emergency. Such phrases, if a patient says ''Please dekh lijiye doctor, I'm feeling very weak'', wherein dekh lijiye means take a look. If these sociolinguistic patterns are not considered adequately by the chatbot, the user intents are misinterpreted causing the major risk and also undervaluing the condition of the patient which might be severe. To tackle a kind of above case effectively, annotated datasets are required for the developers which represents the wide range of context based on cultural, dialects and groups which also enables the chatbot to customize its response to be both appropriate in the contexts and empathetic to the patient.

F. LANGUAGE TRANSCRIPTION AND SPELLING CHANGES

The latin script is used by various users utilizing their native languages such as Hindi, Tamil or Telugu, resulting in the text are transliterated which often lacks spelling that are standardized consistently. For example, lets consider a Hindi word ''dard'' meaning pain can be typed by the patient in different ways like dard, dhard, daard or drrd actually relied on the typing habits of the user and perception of phonetics. This kind of variations will impede the language in the word-level for identification of the language, tokenization and classification of intents and also highlights the processing of noisy and unnormalized data

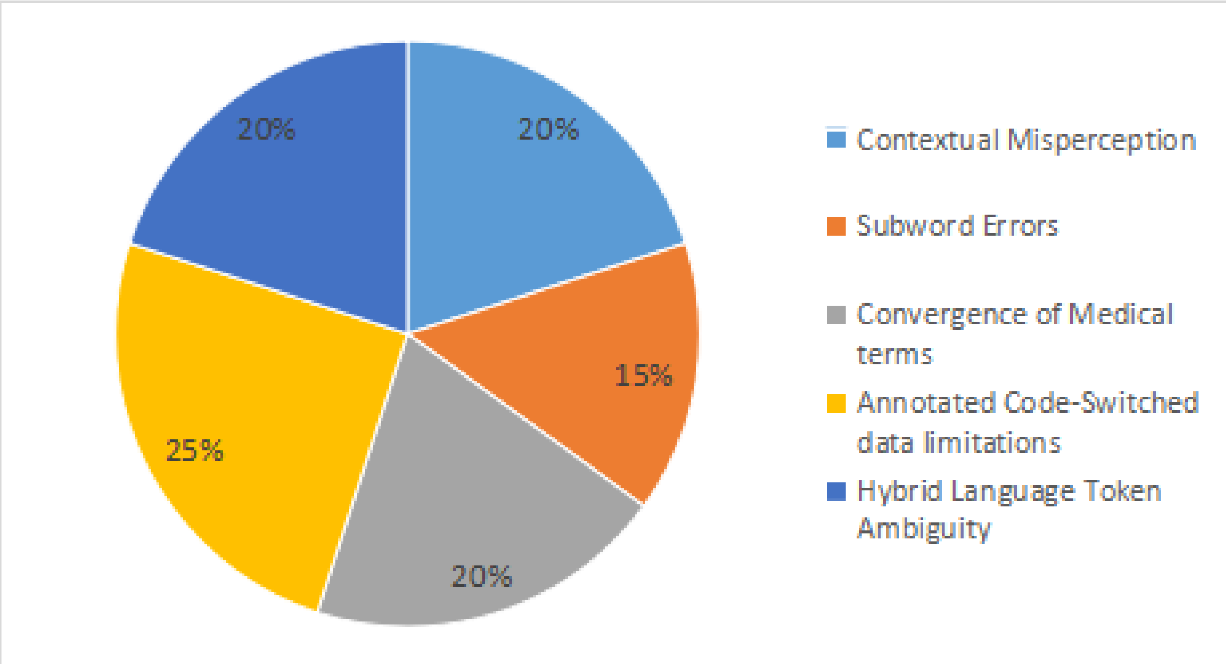

FIGURE 6. Challenges in identification of token-level language.

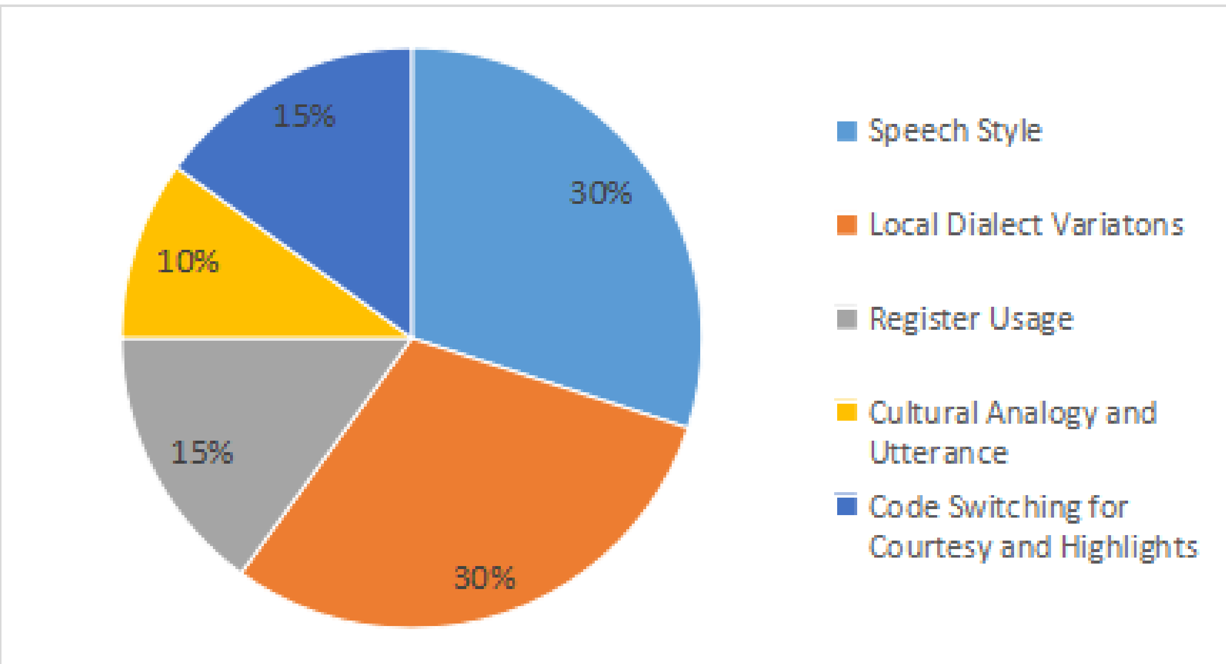

FIGURE 7. Challenges in subject-specific vocabulary.

input effectively towards the requirement for the capability of the system. For the multilingual healthcare chatbot, challenges arise from the transliteration or the language transcription [74] and the spelling changes especially in the environment from where the patients often use the romanized scripts rather than the native scripts. For instance, a patient who speaks Hindi may say, ''Mujhe bukhar hai'' (I have fever) in various spelling like ''bukhaar'' or ''bukkharr''. The challenge here is the spelling variation although they have a similar meaning, so the recognition models of the intent actually depend on the tokens that appear exactly. Comparably, in Tamil or Telugu contexts, patients

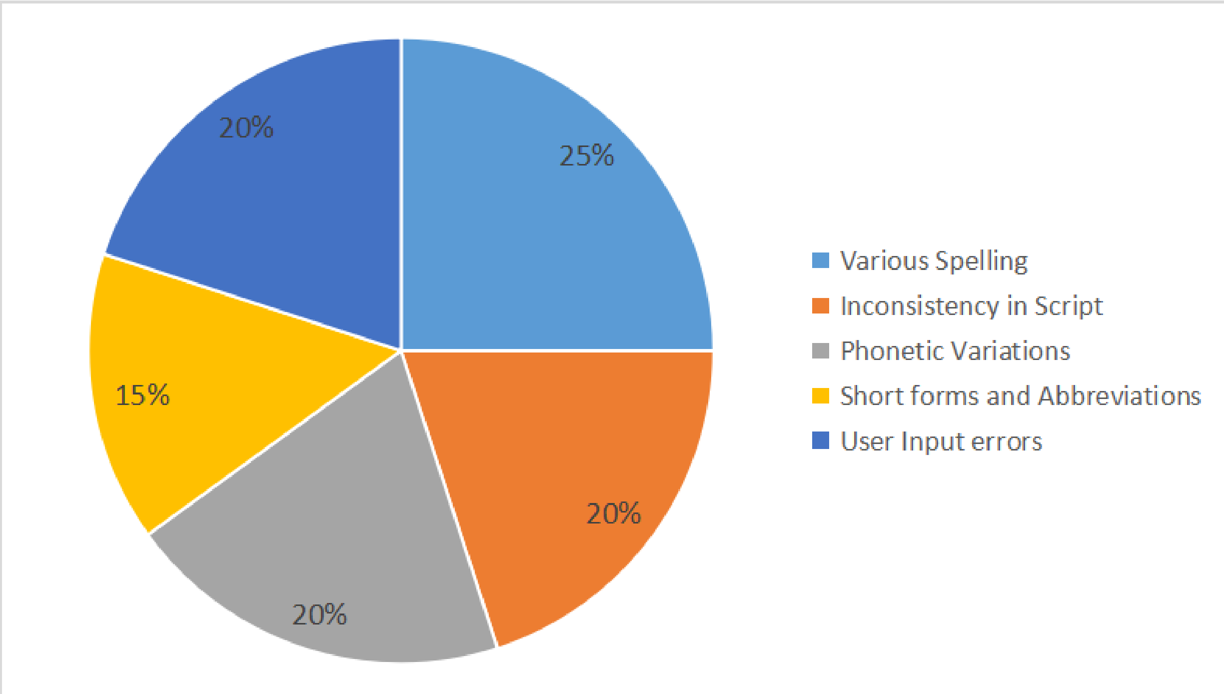

FIGURE 8. **Challenges in language transcription and spelling changes.**

frequently draft the local words in the form Roman script depending on their pronunciation causing inconsistencies like "vayuthan" vs. "Vaiyuthaan" for the gastrointestinal issues. Also few user-generated typos such as 'BP' for Blood Pressure or 'GI' for Gastrointestinal are the abbreviations that further add complexity. Suppose the chatbot is not able to normalize these kind of variations, it causes the major risk in misunderstanding the concerns of the critical health conditions. Figure 8 challenges in language transcription and spelling changes illustrates the spelling changes are caused by various reasons and the language transcription with their appropriate percentile values. To reduce these consequences, researchers make opportunities in the language transcription models, embeddings of sub-words, algorithms that support the normalization of spelling; nevertheless, the annotated transcription or transliteration datasets are led to scarcity in healthcare settings, leading to the accuracy. This case study spotlights the importance of pre-processing pipeline development to manage the inconsistencies in the spelling changes effectively in the communications of real-time multilingual patients.

G. LACK OF REAL-WORLD CONVERSATIONAL DATA

The identification of code-switched languages utilized by most of the datasets is rather artificially generated or extracted from social media platforms. In contrast, the conversations in real-world healthcare settings exhibit the eminent variations in the tone, structural patterns, and the terminology. Anxiety, urgency, or uncertainty can be expressed by the patients in regard to their language, producing expressions with enormous fragments and organized less [75]. The conversations related to healthcare sourced ethically, that are also authentic and anonymized over various languages is essential yet difficult for the concerns of privacy and regulations related to the data protection.

H. EXPECTATIONS OF REAL-TIME PROCESSING

In realtime the healthcare chatbot functions typically to support patients with emergency concerns. In consequence, these models should also strike on the balance between the speed of processing accuracy while complex tasks are executed like identification of languages, detection of intent and generation of appropriate responses. Deployment of edge devices are needed in the lightweight multilingual models. The introduction of a healthcare chatbot in multilinguistic happened in the care of emergency unit to facilitate the patients with quick assessment of symptom and recommendations of triage. There was a query from a patient who was experiencing severe chest pain, which was a mix of Hindi and English language as "Since morning, mujhe chest mein pain ho raha hai, What can I do now" Whereas the chatbot was about to detect the severity of the condition instantly and interpret the input in the mixed language and deliver a response in a real-time basis. Conversely, because of delays occurred in the processing and the translation among the languages, 40 seconds was taken to deliver a response by the chatbot. Meanwhile the patient had acknowledged a call for a emergency service, since the valuable time had

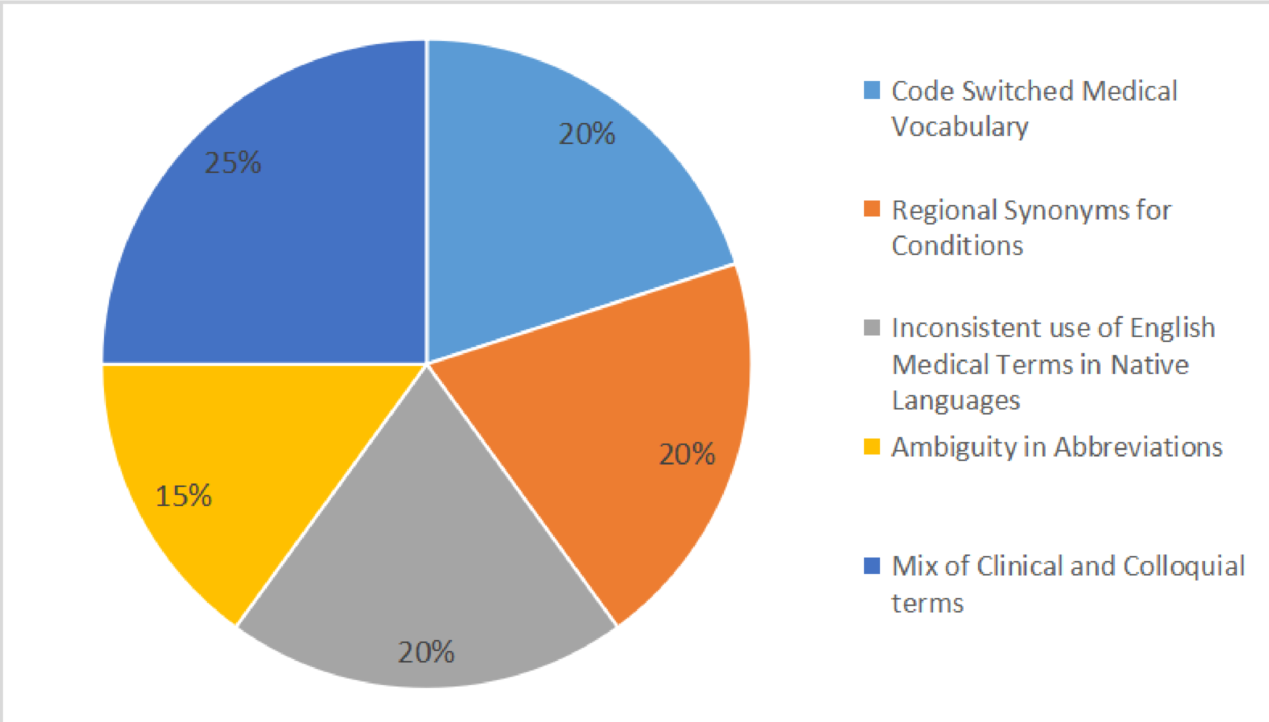

FIGURE 9. Diverse medical terminology.

been lost already. This incident spotlights the patient safety risk produced by the capabilities of real-time insufficient processing in the environments of multilinguistic. The responses with rapidity and precise feedback are expected by the healthcare systems and any delays produced by intensive NLP, detection of language-switching, or connectivity of back-end can compromise critical care of patients. The importance of processing the multilinguistic in real-time is majorly highlighted to manage the emergencies with effectiveness and without delays [61].

I. MANAGING DIVERSE MEDICAL TERMINOLOGY

The terminology of healthcare clinics and Layman expression are blended to form the medical language. The colloquial terms can be expressed by the patients for the symptoms in the context of multilingual, also their native languages along with English words for the critical medical conditions. For instance, words such as "fever", "diabetes", "BP" can be inserted. The subject-specific vocabulary in terms of medicine and the everyday expressions are to be distinguished by the chatbots at the same time, ensuring the understandings consistently over various languages. Besides, expressions like idioms must be combined appropriately to the concepts of medicine namely "gastric discomfort" or "acid reflux". For instance, if a patient as a Hindi speaker says "Mere pet jalan ho raha hai" meaning My Stomach has burning sensation. Effectively managing the medical terminologies which are mixed up with languages remains as an persistent challenge for the healthcare chatbot with multilinguistic, as patients often combine the vocabulary related to colloquial, regional and clinical words in their description or expressions. For instance, an patient in india can say as, "Doctor mujhe BP hai aur Sugar High hai", (meaning doctor I have BP and Sugar). In which the word "Sugar" refers to diabetes and "BP" refers to Blood Pressure. The chatbot must combine these words appropriately to the concepts of healthcare that are standardized, wherein these terms are also colloquially well understood by the chatbot. Likewise, in a context "TB" means Tuberculosis denoted by a abbreviations can be perceived incorrect in a casual speech. The understanding becomes further complicated when its for a regional synonym, for instance "Vayu" is a Tamil word describing the gastric discomfort or usually "gas" as a word in English. These variability results in misclassification of symptoms and misguided with the guidelines of healthcare without the strategies for the robust normalization. In order to overcome and address this, we need to develop a lexicons of subject-specific, embedding of context aware exploitation and combination of knowledge graphs based on the healthcare over the colloquial expressions in terms of healthcare system [76]. The chatbot system in healthcare has to be developed effectively for translation of languages of patients that are informally used and terminologies used by the medical professionals that are precise are highlighted in the above case study. Now the various reasons and their percentile values illustrates to manage the diverse medical terminology used across different languages in Figure 9.

J. DISCREPANCY IN USER AND SYSTEM LANGUAGE

When a chatbot can detect the language of the patient correctly, then the response generation of the similar appropriate language or the dialect turns as another challenge. The flow of conversations in a settings of mixed languages are very difficult to be maintained by the monolingual trained systems or by the parallel corpora [77]. If there are patients who initiate their conversation in their own native language can be felt isolated when the response is entirely in a particular language like English. Conversely, when a regional language has to be switched and may also lead to misunderstandings if the user is unknown or poorly translated with the medical terms. There was a development of the healthcare chatbot in a Multilingual hospital located in India to provide assistance to the patients who are in need of appointments, obtaining advice of the medicines and also details of the prescription were clarified. The need of healthcare chatbots to be developed in such way that can effectively merge the everyday language of the patient with the terminology in the healthcare field are highlighted in the case. Although there were English and Hindi languages majorly developed for the chatbots, still enormous patients interacted in their own regional language(Telugu, Tamil or Kannada) mixed with the language(Hindi or English) in the chatbot. For instance, a sentence was typed by a patient saying ‘‘Telugu lo appointment cheppandi doctor’’ which holds the meaning like ‘‘Please tell the details of the appointment in Telugu. The patient got frustrated since the chatbot wasn't able to do processing in the regional language properly and provided response in the English language only. There were communication barriers created by the misalignment [78], also led to booking of appointment in confusion and medical instructions were also ended up with misunderstandings. In the other example, the instructions of the dosage was conveyed by the chatbot, by default it utilized the English medical terminologies leading to patient misinterpretations. This case brings a spotlight that the support of multilingual without robustness and the handling of code-switching in a adaptive manner reduces the accessibility of the healthcare chatbots specifically to the patients with proficiency limitations in the dominant language of the systems [79].

V. TECHNIQUES USED FOR DETECTING CODE-SWITCHING

A. STATISTICAL AND RULE-BASED APPROACHES

The Statistical and Rule-based methods were considered as the primary techniques on prior how to the advent of deep learning and multilingual transformers and those were utilized for code-switching detection and analyzing part. Today these approaches are intended as traditional, even though these approaches provide beneficial insights and performance with the baseline specifically during the low-resource languages is been dealt with since data scarcity, the deep models could not be practical [80]. In a Multilingual text, tokens or segments are labeled by rule-based and statistical language identification methods, and those methods depend on a limited set of explicit linguistic features. The most widely used criteria are presented below:

- Language Identification over Rule-based: The Methods with the Rule-based models rely on resources from the languages that are predefined like grammar rules, n-grams character and lexicons. These involve the systems typically in detection of script, dictionary lookup and morphological rules and suffix.
- Dictionary Lookup: To identify its similar language. Every token is corresponded to the monolingual dictionaries.
- Detection of Script: The Unicode-based script rules would help in segmentation and classification of rules for any distinct script of languages. For instance, Latin is for English language and Devanagari is for Hindi Language.
- Rule of Suffix and Morphology: In a particular language, common affixes, suffixes or the structures of the morphology are identified by these approaches [81]. For example, English words might contain a silent ‘‘e’’ ending words like ‘‘stage’’, ‘‘face’’ whereas verbs in Hindi frequently end with ‘‘-na’’ like ‘‘sona’’(Sleep), ‘‘khana’’(Food). These rules can be refined in the healthcare chatbots using the subject-specific dictionaries associated with medical terminology to identify keywords like ‘‘BP’’, ‘‘fever’’ or ‘‘diabetes’’, although ingrained in the queries related to code-switching.

The strength of these approaches are considered to be the training data is required less by the rule-based systems and are transparent, also easy for interpreting. With the languages separated clearly, the rule-based systems perform effectively in the controlled environment. Whereas the limitations are in the real-world scenarios, these systems tend to be brittle that also involve in misspellings, transliteration or on the words out-of-vocabulary (OOV) [82]. In addition, the generalization of the dialects across the diverse faces struggle, variations in the user-specific spellings, and the input from the informal chats often encountered in the context of healthcare. The initial approaches relied on dictionaries with language-specific and rules are crafted manually to identify the points of code-switching. Eventually the statistical Models like the Hidden Markov Models (HMMs) were developed for the tags generated sequentially but generalization were offered for limited.

B. MACHINE LEARNING MODELS

Conditional Random Fields(CRFs) and Support Vector Machines(SVMs) are the supervised classifiers which have been utilized for the identification of languages in the token-level [83]. Although, large annotated datasets are required by these models and frequently cannot achieve the language pairs that are unseen. By learning from the data, the Machine Learning (ML) approaches are

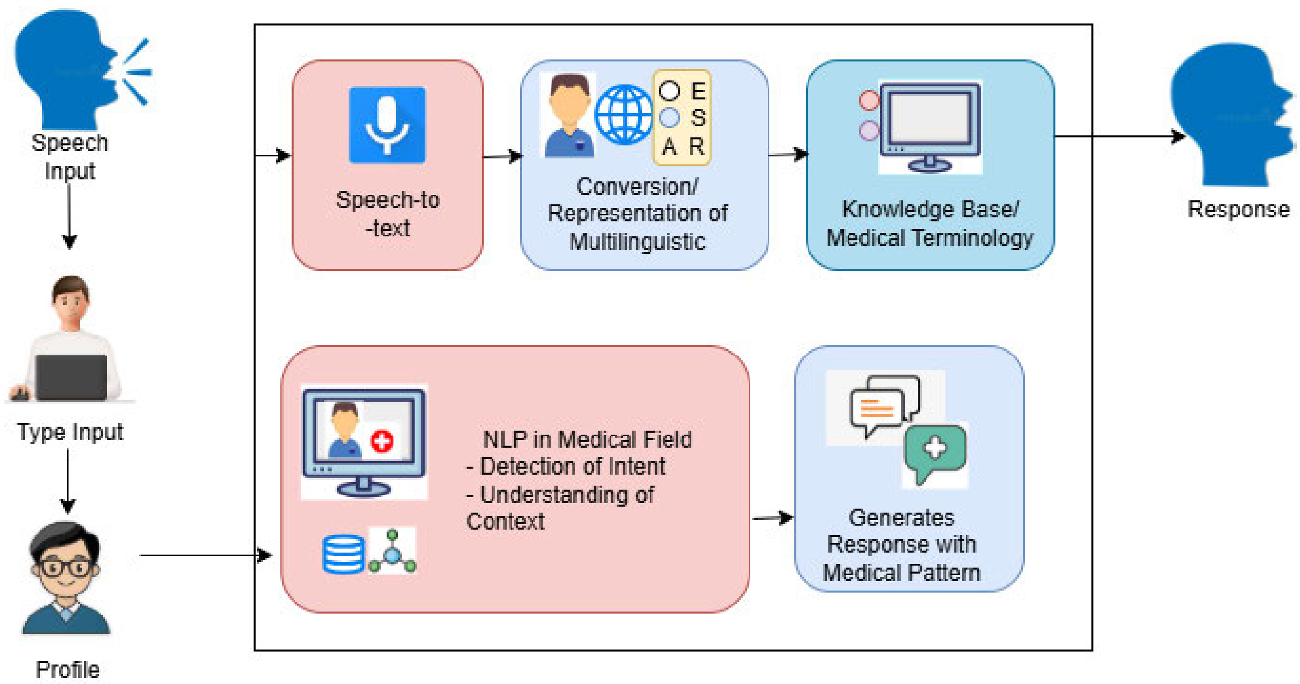

FIGURE 10. Interaction and response generation in multilingual medical chatbot.

considered beneficial in the evolution over the detection of code-switching by exceeding the rigid rules. The detection of statistical regularities in the code-switched data is done by the ability of statistical models like the machine learning which improves the adaptability, scalability and the transferability over the various domain with system with the rule-based [84]. In applications of healthcare, the models are valuable specifically in Machine Learning Models, queries from the users which exhibits the informal language, inconsistent in the phonetics and complexity in the contexts.

1) APPROACHES OF SUPERVISED LEARNING

The SVMs, Decision Trees(DT) and the Random Forests(RF) are considered to be the traditional supervised models utilized for the multilingual input of the classification of intents, identification of languages in the level of token are widely used [85]. Some of the features utilized by the classifiers are like:

- N-grams of character and words.
- Embeddings of the words
- Tags of POS
- Position of the sentence
- Detection of script
- Statistics frequency

SVMs have been used for medical intent detection, like as "reporting of symptoms" or "query based on the medication" from the inputs of code-switching in the domain healthcare. The representation of structure with the hybrid lexical level for those sentences are turned up as a challenge for both languages and the English Language that co-exist incorporated to a window with smaller size of tokens.

2) LABELING SEQUENTIAL WITH CRFs AND BiLSTMs

Token-level language identification are the granular tasks to serve as a baseline with reliability is by the Conditional Random Fields(CRFs) [86]. Moreover, networks with Bidirectional Long Short-Term Memory(BiLSTM) with the deep learning models with the advancements, if merged with the embedding appropriately, illustrates remarkable performance in comparison with the CRF models to be considered as traditional ones. To improve clarity, the individual characteristics of widely used sequence labeling models are as follows:

- CRF Models: Explainable predictions can be provided and also utilizes the features which are hand-crafted. When the data is beneficial only when they are limited.
- BiLSTM Models: These models capture the dependencies with long distance in sequential processing in the directions of both forward and backward, the code-switched data in the context of left and right effective modeling can also improve the accuracy in case of any ambiguity occurs.

Now integration of BiLSTMs along with the CRFs (BiLSTM-CRF models) has validated the effectiveness in different tasks of NLP [82], incorporating Named Entity Recognition (NER) and in the multilingual environments with the language identification.

TABLE 5. Relative analysis on multilingual healthcare chatbots and LLM based systems (2021-2025).

Study (Year)	Model	Languages Supported	Knowledge Base	Evaluation Metrics	Strength	Limitations
PharmaLLM (2024) [94]	Llama-2 tuned with LoRA	High-quality English, Hindi, Bengali	Pharmaceutical DB	Accuracy in F1 Score, Sensitivity-detection rate, Accessibility	Multimodal audio and text integration, extensive language compatibility	Limited low-resource capability
VHC-Bot(2025) [95]	Rasa NER customization	English, Spanish, Arabic	UMLS, SNOMED Clinical Terms	Medical Simulation, Satisfactory	Human-factor design, clinician assistance	Non verbal interface
ChatGPT DBS (2024) [96]	GPT - 4	English, Portuguese	Medical FAQ	DBS Patient Response Monitoring	Bilingual Skill Level	LLM hallucinations Latent
INCET Chatbot (2021) [97]	Support Vector Machine(SVM) + Google Translate	English, Hindi, Gujarati	MedlinePlus	Accuracy	Low computational consumption	Translation inaccuracy

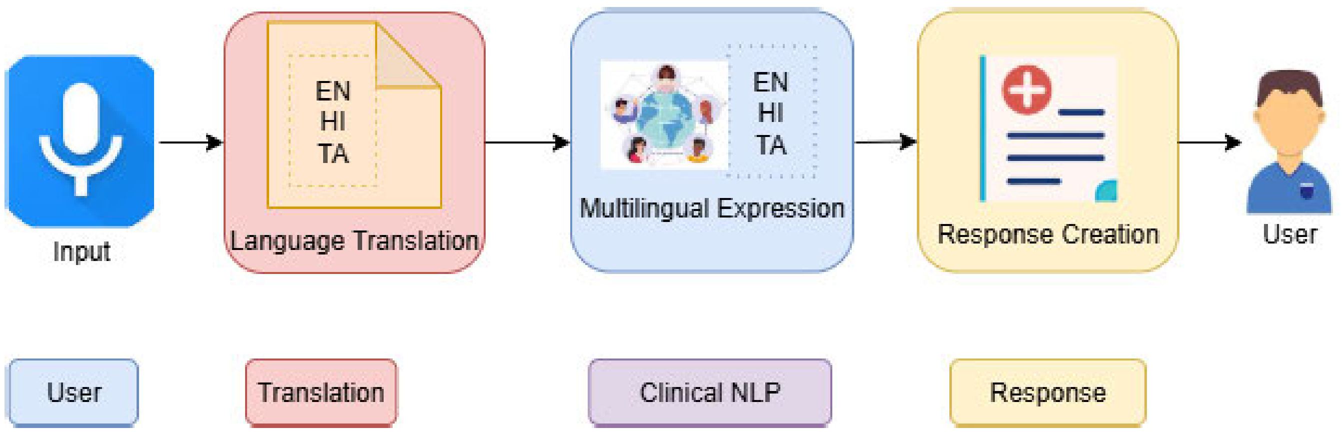

FIGURE 11. Architecture of multilingual healthcare chatbot.

3) STRATEGIES OF EMBEDDING

Many strategies of embedding have been utilized to reflect on the code-switched words. Social Media Data are trained on the Fast-text Embedding (supporting the representations of sub-words). Representations of language aligned to Multilingual Word2Vec vectors. Shared semantic space is mapped with the words from various languages are in the Cross-lingual embeddings [87]. The Hindi-English mixed words like the "pet dard" and the "abdominal pain", which have a similar meaning is are understood by these embeddings enabled by the models, although they are described differently across various languages.

4) HEALTHCARE WITH DOMAIN ADAPTATION

Domain shift is encountered as a challenge in healthcare queries applied by machine learning models that are generic. To address the challenge, techniques with transfer learning are adapted in which the code-switched datasets are initially trained in general consequently refined over the healthcare-specific data. Paraphrasing and back-translation methods of data augmentation are developed to produce the synthetic multilingual queries in healthcare are for improving the performance of the model [88].

C. MODELS OF NEURAL AND TRANSFORMER NETWORKS

The models that are transformer-based, like mBERT (Mulitlingual BERT), MuRIL (Multilingual Representations for Languages of India), and XLM-R (XLM-RoBERTa), are at present high-tech pioneering [89]. They can improve the accuracy in the refined code-switched datasets and also manage the tasks of token classifications. MuRIL, for example performance of the impressive languages like English-Hindi and English-Tamil data is mixed, have be illustrated [90]. The multilingual corpora are large, over over-pre-training provides models with the leverage in the generalization and contextual understanding. The inception of models from deep learning and transformer-based models has transformed Multilingual NLP drastically with the multilinguistic models, enabling nuanced understanding and more sophisticated interpretations of the text with code-switch. The models have illustrated effectiveness exceptionally in crucial fields like healthcare, in which the context is understood with an ability, achieving the alignment of cross-lingual and in various languages, its generalization functions as a major contribution in improving the safety of a patient and the chatbot dependency. In Table 5, Relative Analysis on Multilingual Healthcare chatbots and LLM based systems from the year 2021 to 2025 are analyzed.

1) EVOLUTION AND VARIANTS OF RECURRENT NEURAL NETWORKS (RNNs)

Initially, the approaches of the neural networks for the detection of code-switching were used by the Recurrent Neural Networks(RNNs) and their variants, with advancements in models such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) [95]. Especially for the sequence labeling task, these models were specific to these networks, having the language identification for the networks at the token level. For instance, the conversational data in the form of Hindi-English is trained by the BiLSTM may label the sequence more effectively. The learning of contextual dependencies outperforms with the BiLSTM models, but finds it difficult to manage the relationship that lasts over long range and needs substantial annotated data for effective results.

2) DEVELOPMENT OF TRANSFORMER-BASED MODELS

BERT model, as a transformer-based model and its variants supporting multilingual, signified a significant transformation in the abilities of NLP. The mechanisms supporting self-attention are mostly dependent on these models to approach the sequence as a whole and in parallel. This approach can enable the performance of the inputs with code-switching and Multilingual for a better results. Some of the conspicuous models utilized in the context can include:

- mBERT(multilingual BERT): The model is trained on 104 languages with a collective lexicon and dimensional input space. The Task of token-level classification is effective, although it still lacks particular adaptations to the inputs supporting code-switching.
- XLM-R (XLM-RoBERTa): The model is trained stronger with Multilingual over 2.5 TB of common web crawl dataset spanning 100 languages. It furnishes an upgraded performance over mBERT, specifically in the tasks of healthcare in downstream, incorporating noisy and informal text.
- MuRIL (Multilingual Representations for Languages of India): The model is built by Google, which is especially pre-trained over languages of India, embracing mixed languages with code-switching such as Hindi-English or Tamil-English. It operates the expressions dealing with transliteration, named entities and subject-specific eclipsing the multilingual models in general.

Figure 10 explains the interaction and the generation of responses in the multilingual medical chatbot, and the architectures of transformer models in the multilingual healthcare chatbots are utilized. These models are also put into the tweaking or purpose of calibration in token classification (for language identification) [70], classification of sequences (for detection of intents), and the generation of responses (through the frameworks of encoder-decoder). Transformer-based models have become an integral part of modern healthcare chatbots due to their ability to model long-range contextual dependencies. These architectures are utilized along with the following core NLP components within the NLP pipeline:

- Language Token Identification: For instance, in a mixed input with various languages, identifying the Hindi or English words [96].
- Patient Intent Classification: Identifying if a patient is reporting any symptom, seeking for a remedy or appointment.
- Multilingual Response Generation: Generating an fluent, contextually relevant responses that can manage the accuracy and the precision medically.

These transformer-based tasks facilitate the functional objectives of healthcare chatbots as presented below.

- Assessment of Symptom - Language-specific symptom evaluator
- Guidelines of Medication - Guidance on drug dosage and conversations
- Support for Chronic disease - Assistance in bilingual for the afflictions of diabetes and hypertension.
- Aid to Mental Health - Supporting emotional care with respect to the cultural difference over various languages.
- Emergency Recommendation - Practical environment, multilingual first-aid, and crisis directions.

For example, there can be a response generated by a fine-tuned mT5 model (T5, Multilingual version) in a mixed query of Hindi-English for Hindi. Despite their advantages, neural network models experience many challenges:

- Availability of data with Limitations: There is an annotated healthcare dataset, which is with a scarcity of large-scale data along with code-switching, confining the fine-tuning of the model.
- High Computational demands: Substantial processing power is required by the transformer model, which regulates its practicality with the constraints of resources in the environment.
- Equity Concerns and Bias: Societal and cultural bias are inherited by the pretrained models, which results in biased or inappropriate retorts in the context of healthcare.

VI. RETORTING TO MULTILINGUAL QUERIES

A. DETECTION OF INTENTS

The languages involved are identified and deriving the information that are related and also the subsequent step is to understand the intent of the patient. This has an inclusion of

- Reporting the symptoms
- Requests for appointment
- Managing the emergency inquiries
- Responding to the questions related to medication and diet.

Detecting the intents in the scenarios of Multilingual contexts needs the training of models [97] over the intents from various languages and expressions in diversity.

B. STRATEGIES FOR RESPONSE GENERATION

There are two primary methodologies for approaching. Template-driven are the recognized intent is a major cause for selecting the predefined responses across languages. Neural Synthesis are the Encoder-decoder frameworks are utilized for the creation of complete response. Such frameworks are mBERT and mT5. Reinforcement learning is used for training the Multilingual transformers and the context of conversations creates more contextually appropriate and natural responses.

C. RETENTION OF CONTEXT

Frequently the healthcare interactions inculcate many exchanges, preserving the coherence of linguistic across the conversations with code-switching. The inclusion of strategies such as:

- Handling the dialogue states enclosed in a multilingual framework.
- Making use of memory from cross-lingual networks
- Medical chatbot in the Mulitlingual designs

Systems with Pipeline centric models have the Configuration with the partitions grants single language resources to be re-utilized but there are errors produced in the translation which creates inaccuracy in the clinical scenarios. Multilingual Model comprehensive across various languages, the processing of input and the generation of output are handled by these frameworks although there is a demand for Multilingual medical data for training. Retrieval Enhanced Generation (RAG), the approach is considered as hybrid and utilizes the generalized framework models assuring the accuracy [107], informative and appropriate responses in the contexts, whereas framework is directed by the selected experts. In Table 6 the various dataset used with various combination of languages in the specific domain are explored.

VII. ETHICAL ISSUES AND OBJECTIVITY BASED ON THE NLP

A. LANGUAGE-BASED MODEL BIAS

The prejudices present in the training datasets may intensify the Multilingual models, which results in responses with unsuitability or discriminatory, particularly in the fields which are sensitive like healthcare. The Essential part is the validation of impartiality over various languages and dialects.

B. RISK OF INAPPROPRIATE DIAGNOSIS

The harmful misinterpretations are caused by the errors in the interpretation of code-switched queries. Chatbots should be developed to seek the clarification for input that are ambiguous and elevate the serious conditions to the qualified professionals in the healthcare service.

C. DATA MANAGEMENT AND CONFIDENTIALITY

Sensitive information of the patient can be managed with the utmost diligence. Data Processing in cross-language should adhere to the regulations like HIPAA (USA) or DPDP (India) [108].

VIII. INNOVATIONS IN MULTILINGUAL HEALTHCARE

A. OBJECTIVES FOR CODE-SWITCHED DATA PRETRAINING

The outcomes of multilingual applications are enhanced by specifically capturing of code-switching patterns by creating a novel pretraining task.

B. INTEGRATION OF MULTIMODAL

The comprehension of symptoms in various languages is enhanced by integrating input spoken in the form of textual data from audio and visual data, like the reports that are scanned copy [109].

C. CUSTOMIZATION

For natural and effective interactions, the chatbot should adapt to the preferred language of the patient, medical history, and health conditions.

D. ANNOTATION PLATFORMS BASED ON COMMUNITY DRIVE

The expansion of available relevant datasets can be rapidly done by leveraging the efforts related to crowd-based, which involves the annotation of code-switched dialogs in healthcare.

IX. MULTILINGUAL NLP

In recent years, multilingual NLP has been considerably advanced in the field of healthcare chatbots. A major challenge considered in research is code-switching, which involves handling patient queries [110]. Addressing the issue requires improvement in language identification, classification of intents, and response generation with context sensitivity, especially incorporated in low-resource and real-world environments.

From the above Figure 11, the multilingual healthcare chatbot architecture is illustrated with the input and the response generated from the chatbot using the translation and multilingual expression. The research outlines serve as a road map and chatbots that are designed with robustness, inclusivity, and ethics, also capable of providing to the linguistic population across diversity. Multilingual systems play a vital role by integrating subject-specific datasets,

TABLE 6. Dataset for multilingual speech and text.

Dataset	Year	Language Pair(s)	Domain	Accessibility
SEAME [98]	2010	Mandarin-English	Speech	Public (on request)
MUCS [99]	2021	Hindi-English, Bengali-English	Speech	Public
TALCS [100]	2020	Mandarin-English	Speech	Public
ArzEn [101]	2024	Egyptian Arabic-English	Speech	Public
ESCWA [102]	2024	Arabic-English, Arabic-French	Speech	Public
Miami Bangor	2014	Spanish-English	Speech	Public
LinCE [103]	2023	Spanish-English, Hindi-English, Nepali-English, Arabic	Text/NLP	Public
CroCoSum [104]	2024	Various (code-switching)	Summarization/Text	Public
CAFE [105]	2023	Algerian Arabic-French-English	Speech/Text	Public
MSCS [106]	2024	Mandarin-English	Speech	Public

context-based models, and AI practices with responsibility for ensuring accessibility in equivalence.

X. CONCLUSION

The evolution of multilingual and code-switched healthcare chatbots has great potential to overcome language barriers and enhance access to quality healthcare information. Facing challenges such as linguistic diversity, low-resource languages, and contextual complexity, research into language detection, multilingual embeddings, and transformer-based models is paving the way for more effective and accessible conversational agents. A Sustained effort to create effective evaluation measures and to respond to ethical concerns will be essential to ensure that such technologies enhance healthcare equity and empower heterogeneous communities. This review identifies several unresolved challenges, including the limited availability of healthcare-specific multilingual datasets and persistent errors in code-switched understanding. Performance gaps remain evident in low-resource and real-world settings, particularly for medical entity recognition and intent classification. Addressing these gaps requires future research on robust multilingual modeling, domain-adaptive training, and safety-aware deployment strategies.

By synthesizing existing research and highlighting open challenges, this review aims to guide future efforts toward the development of equitable, trustworthy, and linguistically inclusive medical chatbots capable of truly democratizing AI-driven healthcare support. Ultimately, multilingual healthcare chatbots can revolutionize patient engagement and care, making healthcare more accessible and tailored to the needs of multilingual communities.

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